

EXPERIMENT-17 Sum Integers from 1 to N Prolog

Aim

To write a **Prolog program to calculate the sum of integers from 1 to N.**

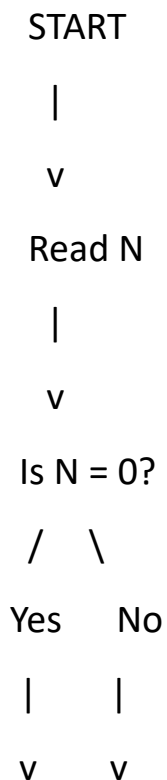
Objectives

- To understand recursion in Prolog.
- To implement arithmetic operations.
- To calculate the sum from 1 to N.

Algorithm

1. Read N.
2. If $N = 0$, return sum as 0.
3. Otherwise, recursively calculate the sum up to $N-1$.
4. Add N to the previous sum.
5. Display the result.

Flowchart



Sum = 0 Sum(N-1)

1

V

Add N

1

V

Display Sum

1

V

STOP

Prolog Program

% Sum of integers from 1 to N

sum_to_n(0, 0).

sum_to_n(N, Sum) :-

$$N > 0,$$

N1 is $N - 1$,

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sum_to_n(N1, Sum1),
```

Sum is $N + \text{Sum1}$.

Query

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?- sum_to_n(5, S).
```

Output

[illegible]

Result

The sum of integers from 1 to 5 is **15**.

Conclusion

Thus, a recursive Prolog program was successfully implemented to calculate the sum of integers from 1 to N.